Band-1 Hadamard Order 9092 – Classical Paley-I (q=9091); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9092 Paley-I PASS; Paley-II SKIP (q=4545=3^2*5*101 NOT prime power); Kronecker FAIL (4*2273); skip 9076 honest; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-21_Band1_Hadamard_8948_Miyamoto`; `2026-09-21_Band1_Hadamard_8884_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_9092_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9092 only (verified max|HH^T-nI|=0), via classical Paley-I q=9091 (prime ===3 mod 4). 9092 IN SPK via Paley-I.** Honest skip: **9076** this wave (Miyamoto-only expected; seed Paley-II $q = 1133$ composite). Classical SPK for 9092: Paley-I **PASS**s ($q = 9091$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL** ($q = 4545 = 3^2 \cdot 5 \cdot 101$ NOT prime power); Kronecker **FAIL**s (4×2273; also Kronecker(4×568)=$2272 \neq 9092$ – H568 integrity only). **9092** folds into SPK **only** via classical Paley-I. It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **2220+8884+2236+2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **9092**.

- H_{9092} = Paley-I ($q = 9091$) – classical SPK
- Paley-II for 9092: **SKIP** ($q = 4545 = 3^2 \cdot 5 \cdot 101$ NOT prime power)
- Kronecker for 9092: **SKIP** ($4 \cdot 2273$ not admissible; $4 \times 568 = 2272 \neq 9092$)
- H568 integrity (optional): PASS (seed file only)
- Skip: **9076** (Miyamoto-only expected)

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_9092.py`

Results. overall_pass=True; units 91/91; wall 610.8087s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9092 = q + 1$ with $q = 9091$ prime $\equiv 3 \pmod 4$ -> Paley-I eligible.
- Paley-II: $q = 4545 = 3^2 \cdot 5 \cdot 101$ NOT prime power -> SKIP.
- Kronecker: $9092 = 4 \cdot 2273$; 2273 prime $\equiv 1 \pmod 4$ not Hadamard-admissible -> FAIL.
- Note: Kronecker(4×568)=2272 $\neq$ 9092 (H568 npz integrity only).
- Prior optional skipped: 9076 (Miyamoto-only expected).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9092 Paley-I	Paley-I q=9091	0	603.2197
H_9092 Paley-II	SKIP q=4545=3ˆ2*5*101 NOT prime power	–	–
H_9092 Kronecker	SKIP 4*2273 not admissible	–	–
H568 integrity	load npz (!=9092 path)	0	0.1022

SPK closure note

Seeds include prior SPK through **8884** lineage (**2220+8884**; **2236**; **2160**; **2140+8564**; **1064+2128**; ...; **NOT** 8948/8644/8516/...) + verified **9092**. Next constructive gap **2052** (OPEN).

Quantity	Value
9092 in SPK	**True** (Paley-I q=9091)
Gap 2052	**OPEN**

Quantity	Value
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=2281->n=9124 Paley-II(q=4561) triage – q=4561 prime ===1 mod4 -> classical SPK candidate OR q=2293->n=9172 triage – Paley-I q=9171 composite; Paley-II q=4585=5 7131 NOT prime power OR q=2276->n=9104 Paley-I q=9103 prime ===3 mod4 – alternate SPK OR q=2282->n=9128 Paley-I q=9127 prime ===3 mod4 – alternate SPK OR method-attack SPK gap 2052 OR Track B / Object 1 meter await